

FLIGHT STATUS PREDICTOR

Predicting if a flight will arrive at least 15 minutes later
than scheduled.

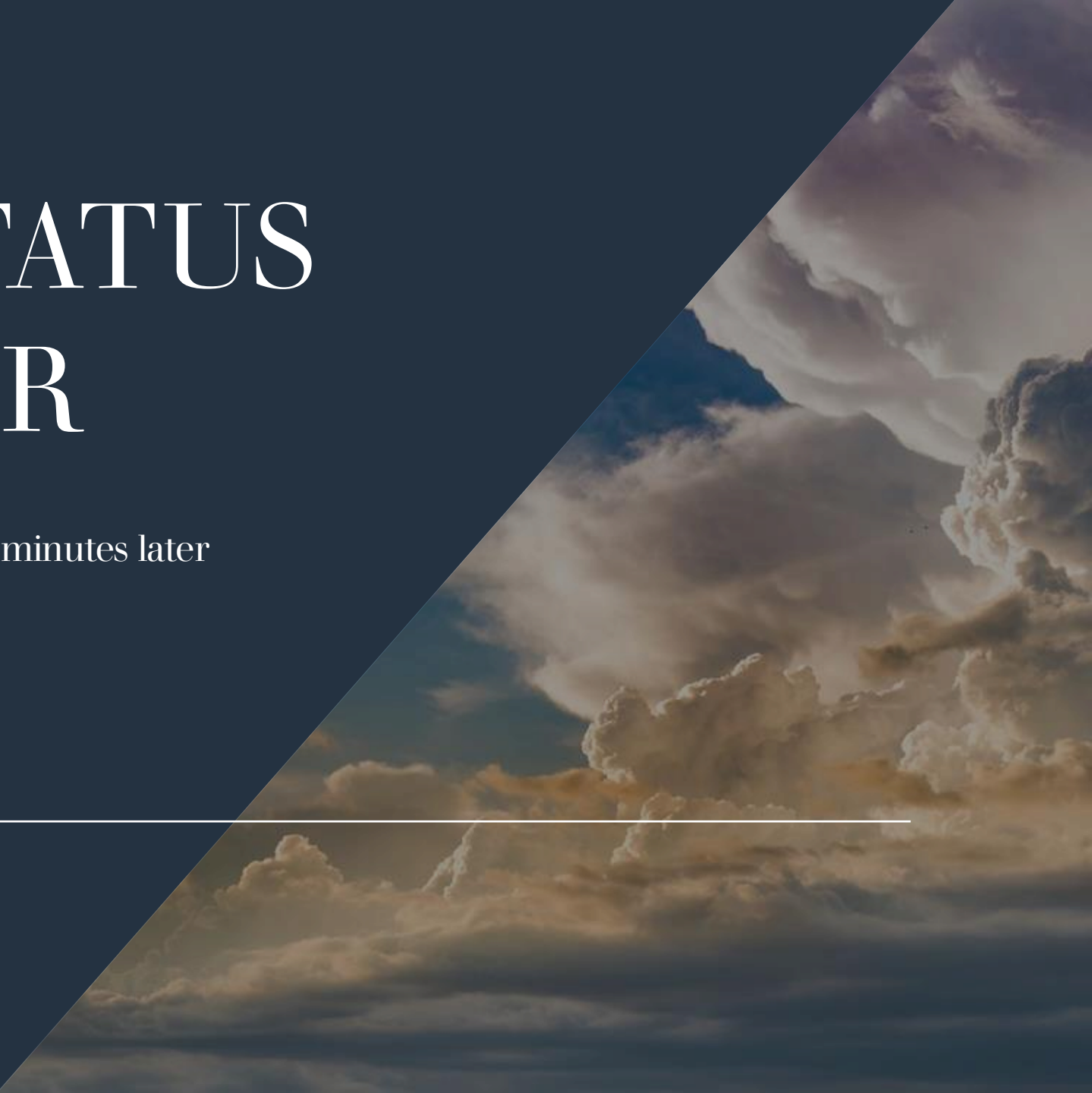


TABLE OF CONTENTS

- Introduction
- Overview
- Data
- Methodology
- Results
- Conclusion

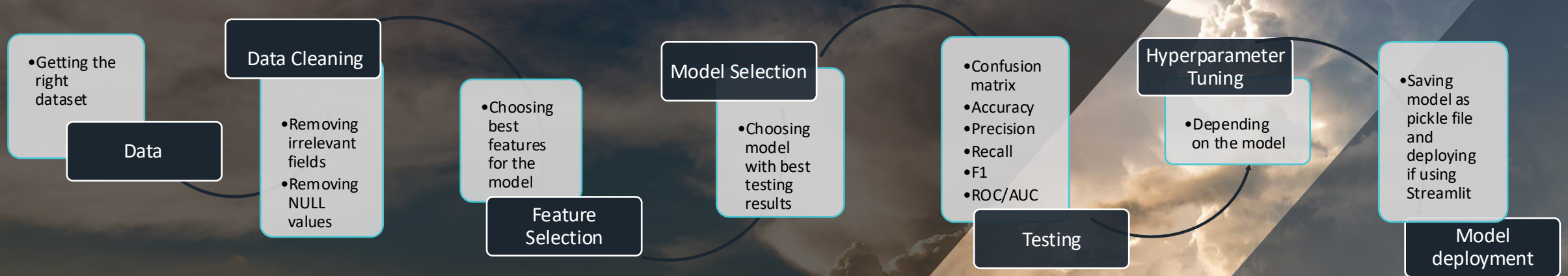
PROBLEM

- Goal: Predicting flight delay
- Solution:
Use ML classification methods
- Independent variables –
- Response variable:
 - Arr_Delay_Minutes – regression problem
 - Arrival_Delay_At_least_15_Minutes – classification problem

OVERVIEW

- Dataset acquisition
- Delay problem – predicting if a plane will land with at least 15 minutes of delay.
- Feature engineering – choosing the best features
- Choosing machine learning methods to be implemented
- Choosing the best models

PROCESS

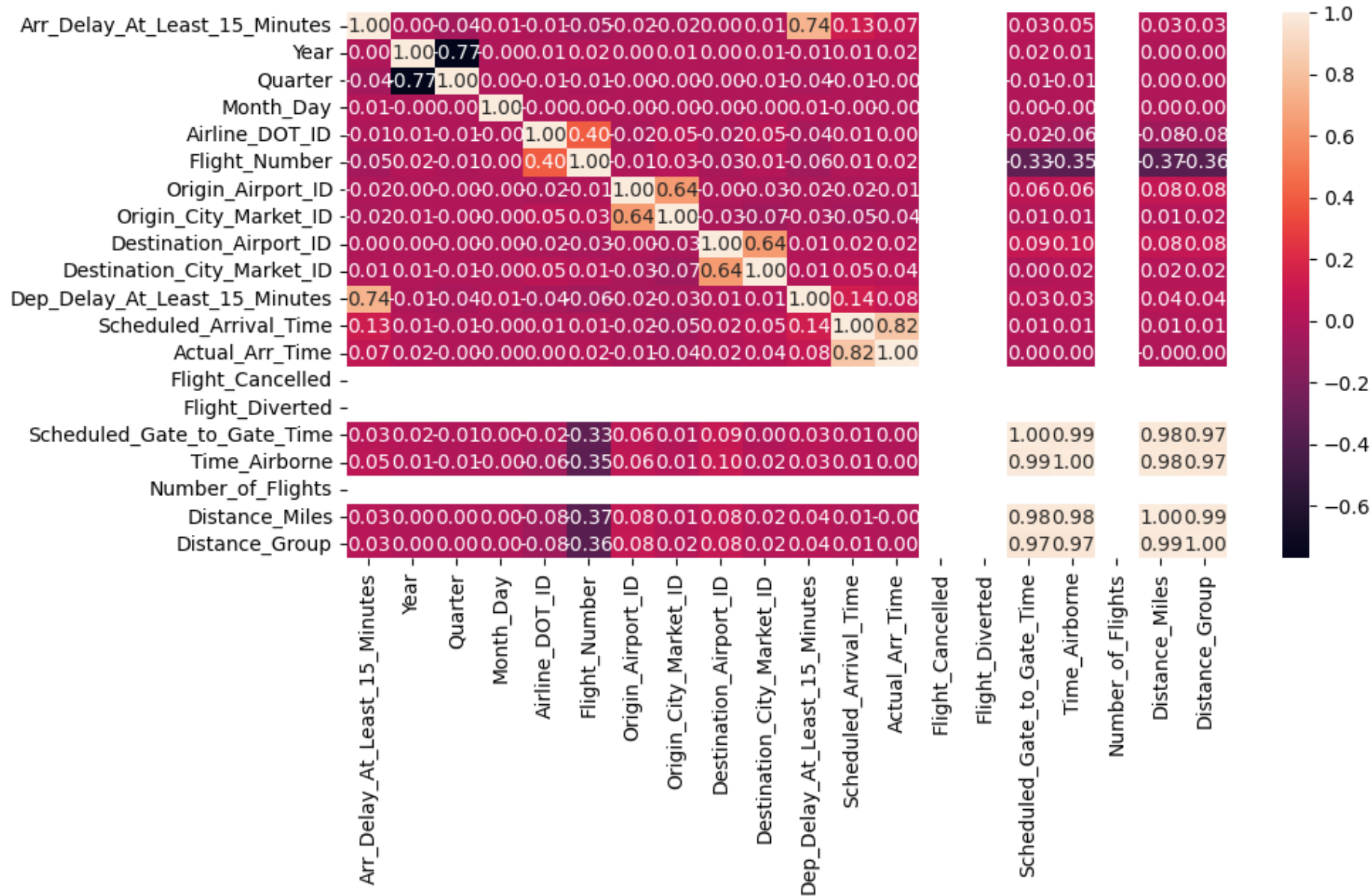


DATA

Original Data Set :

- 6,777,978 rows and 43 columns (11 categorical and 32 continuous)
- 15 min delay probability 20.2 %
- After transformation (removing certain features and rows)
 - 2,742,902 rows, 27 columns
 - 15 min delay probability 50 %
- Columns with NULL values were removed.

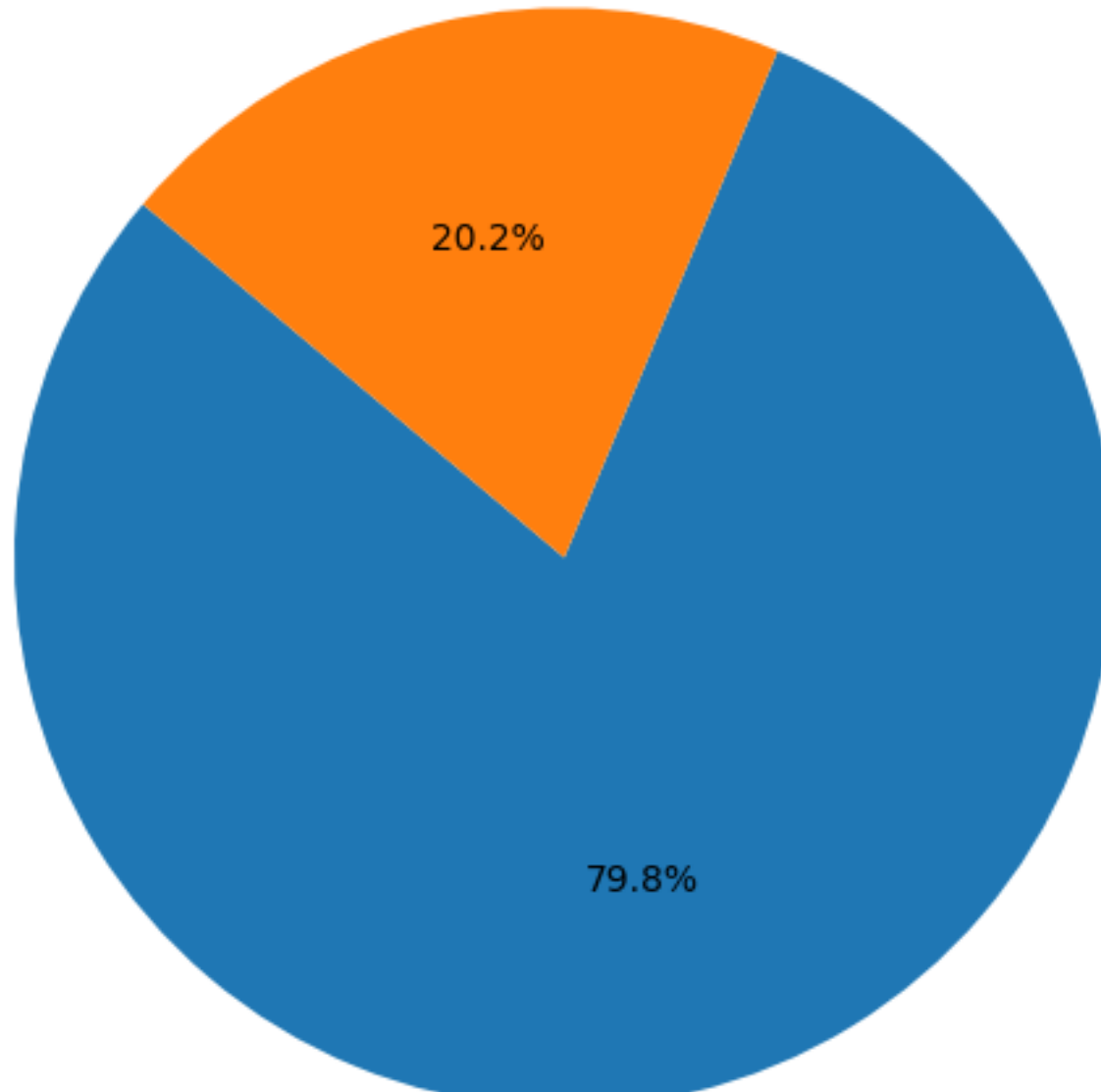
DATA – CORRELATIONS BETWEEN FEATURES



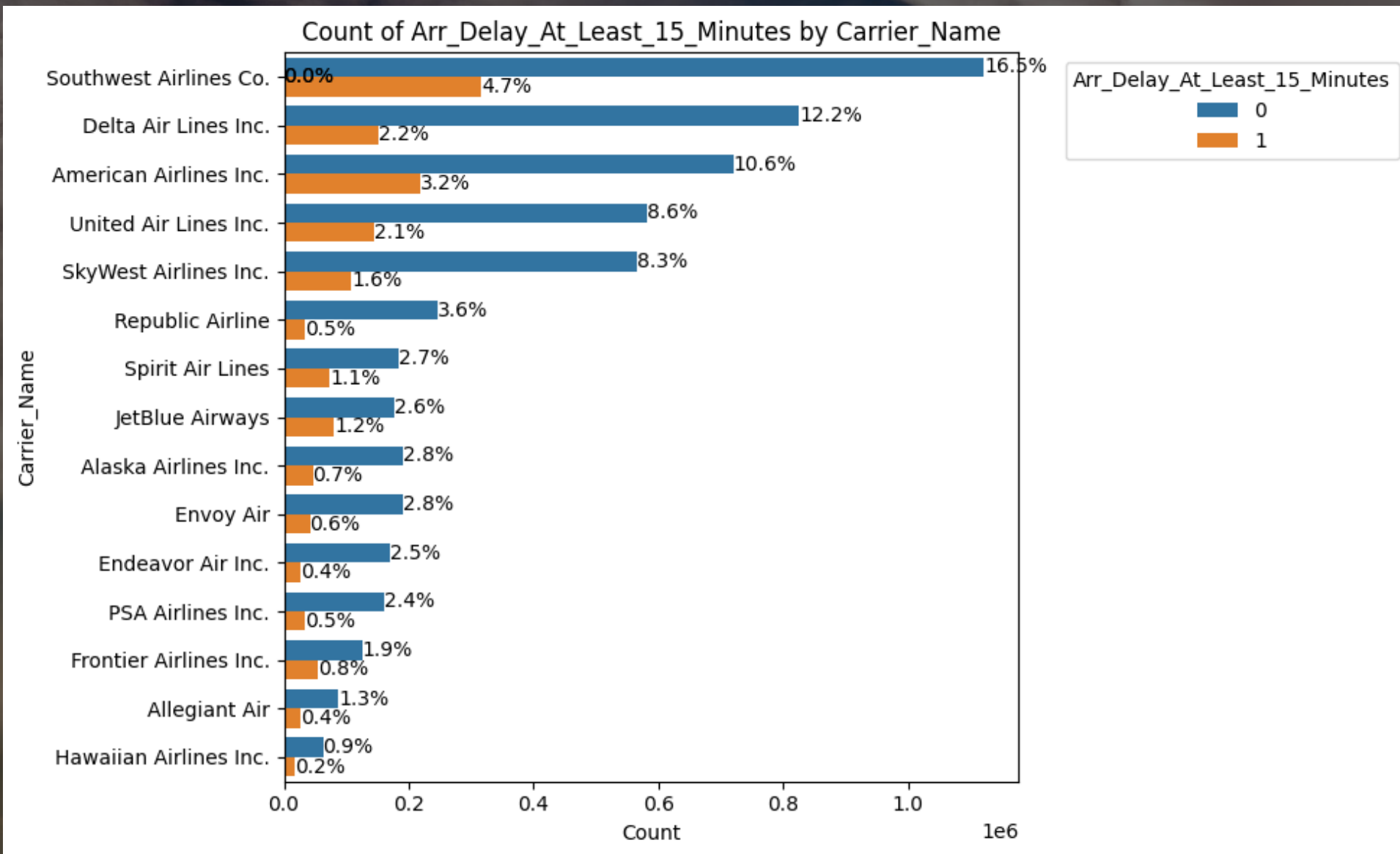
PRELIMINARY DATA ANALYSIS

Distribution of Arrivals with Delay At Least 15 Minutes

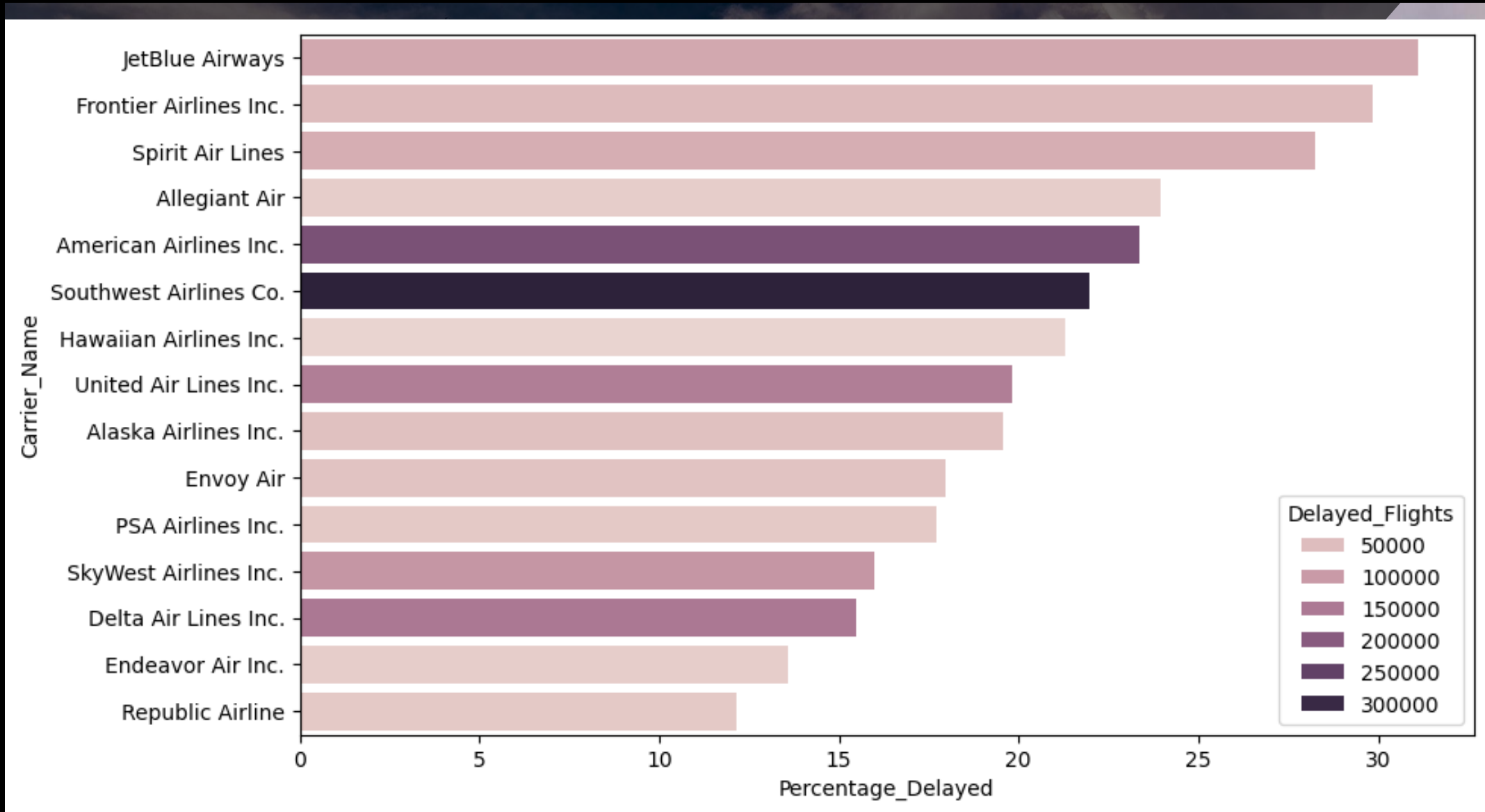
1



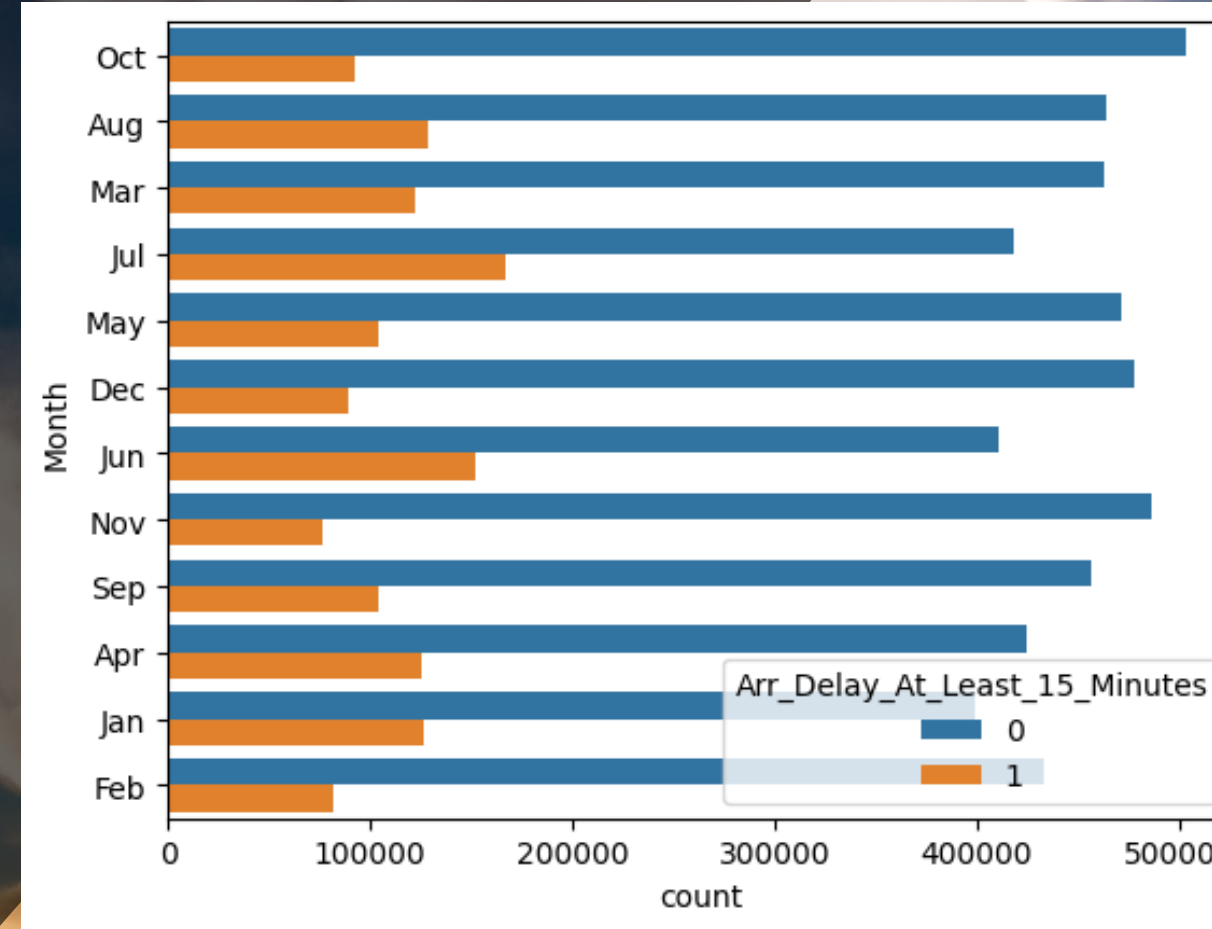
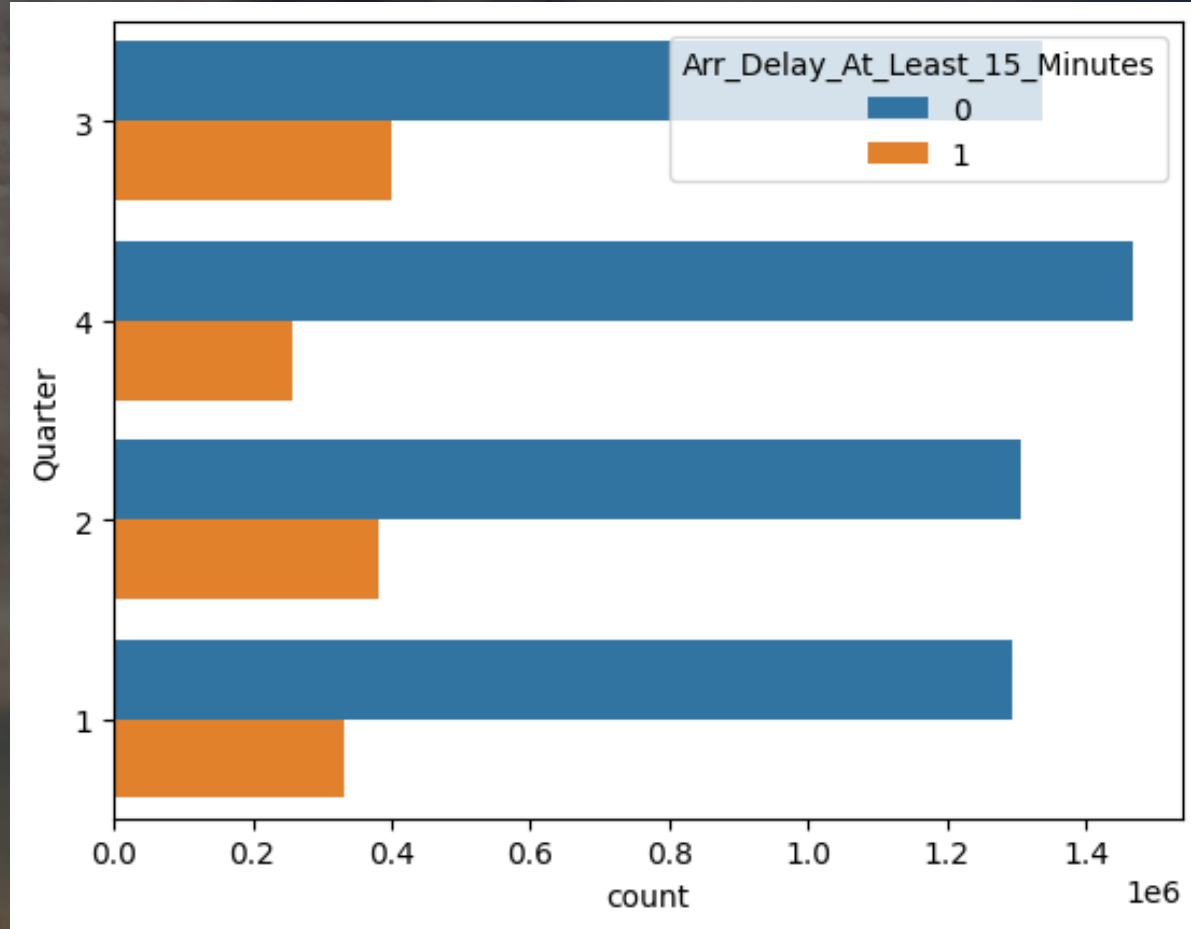
EXPLORATORY DATA ANALYSIS



EXPLORATORY DATA ANALYSIS

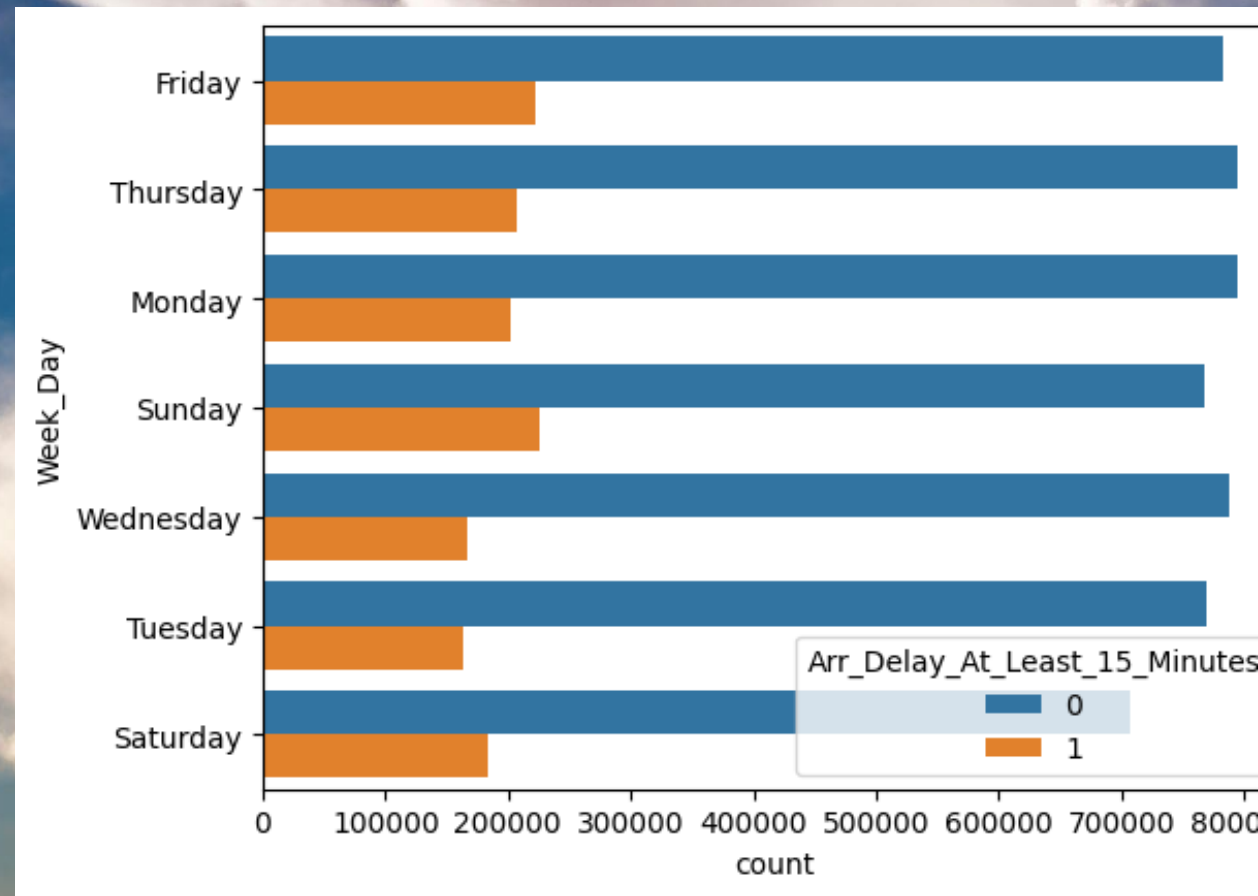


EXPLORATORY DATA ANALYSIS



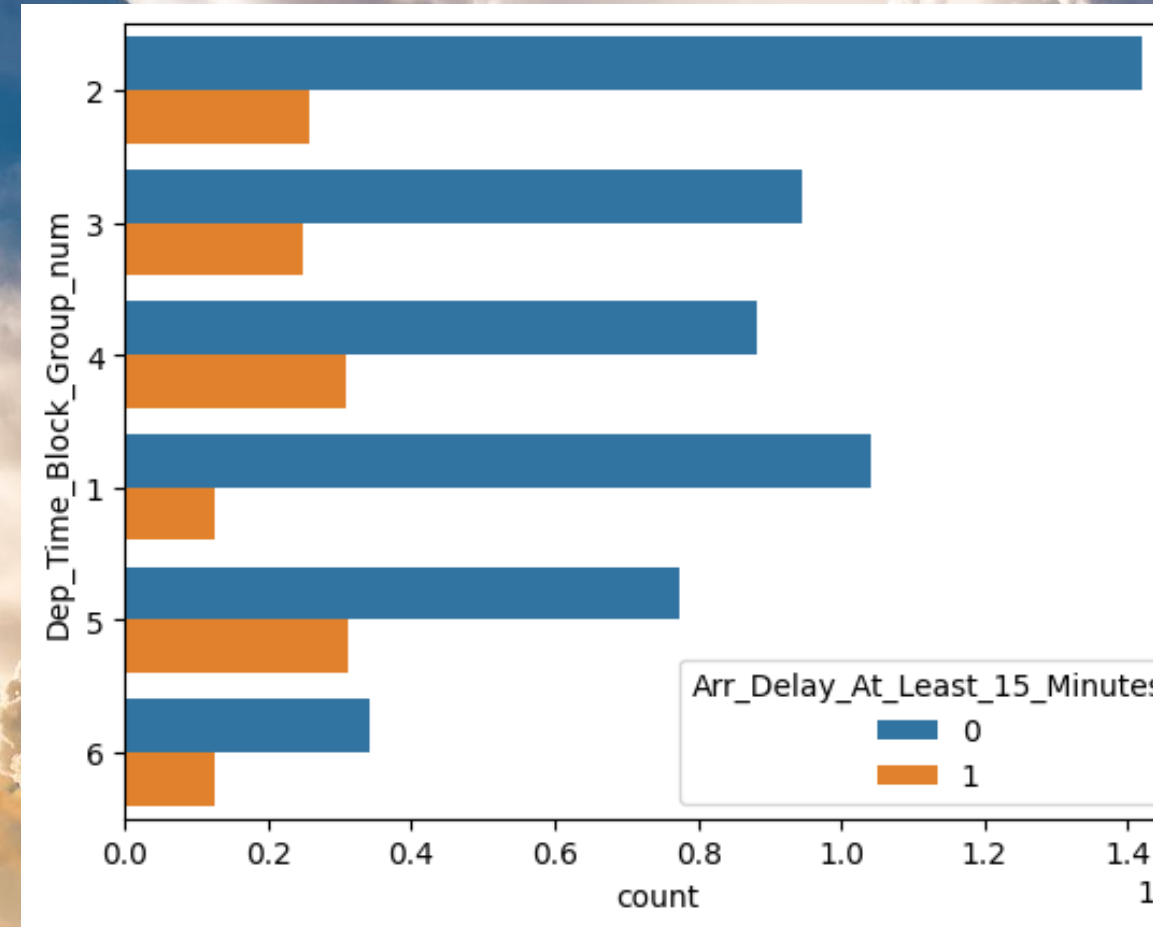
EXPLORATORY DATA ANALYSIS

Feature_class	Total_Flights	Delayed_Flights	Percentage_Delayed
Sunday	993,917	225,513	22.69%
Friday	1,006,397	222,892	22.15%
Thursday	1,002,419	207,300	20.68%
Saturday	890,364	183,081	20.56%
Monday	996,731	201,586	20.22%
Tuesday	954,067	164,199	17.58%
Wednesday	954,083	166,880	17.49%



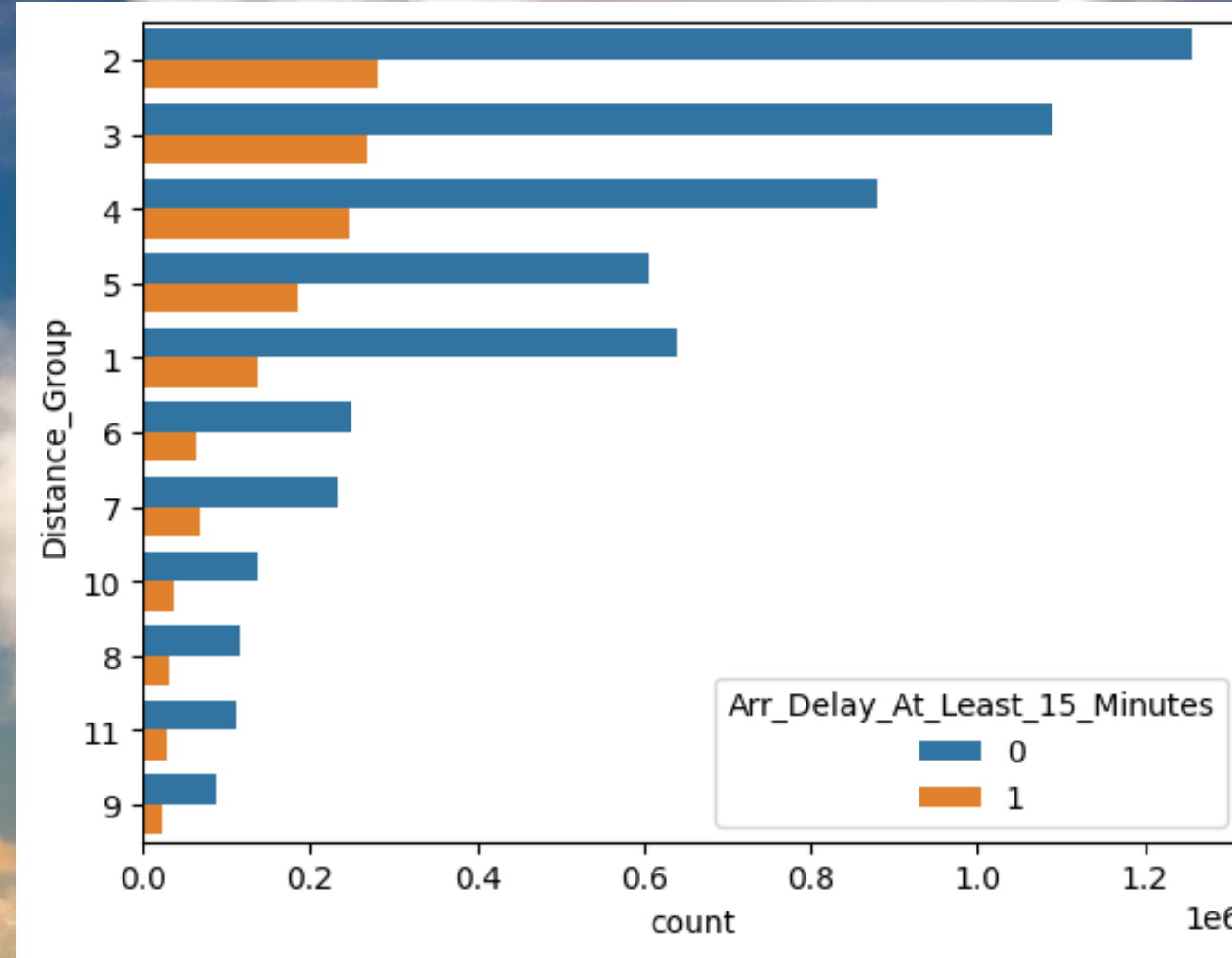
EXPLORATORY DATA ANALYSIS

	Total_Flights	Delayed_Flights	Percentage_Delayed
Evening	1,085,963	311,197	28.66%
Night	467,796	125,100	26.74%
Afternoon	1,190,516	307,259	25.81%
Early afternoon	1,193,596	247,230	20.71%
Morning	1,675,136	256,402	15.31%
Early morning	1,164,971	124,263	10.67%

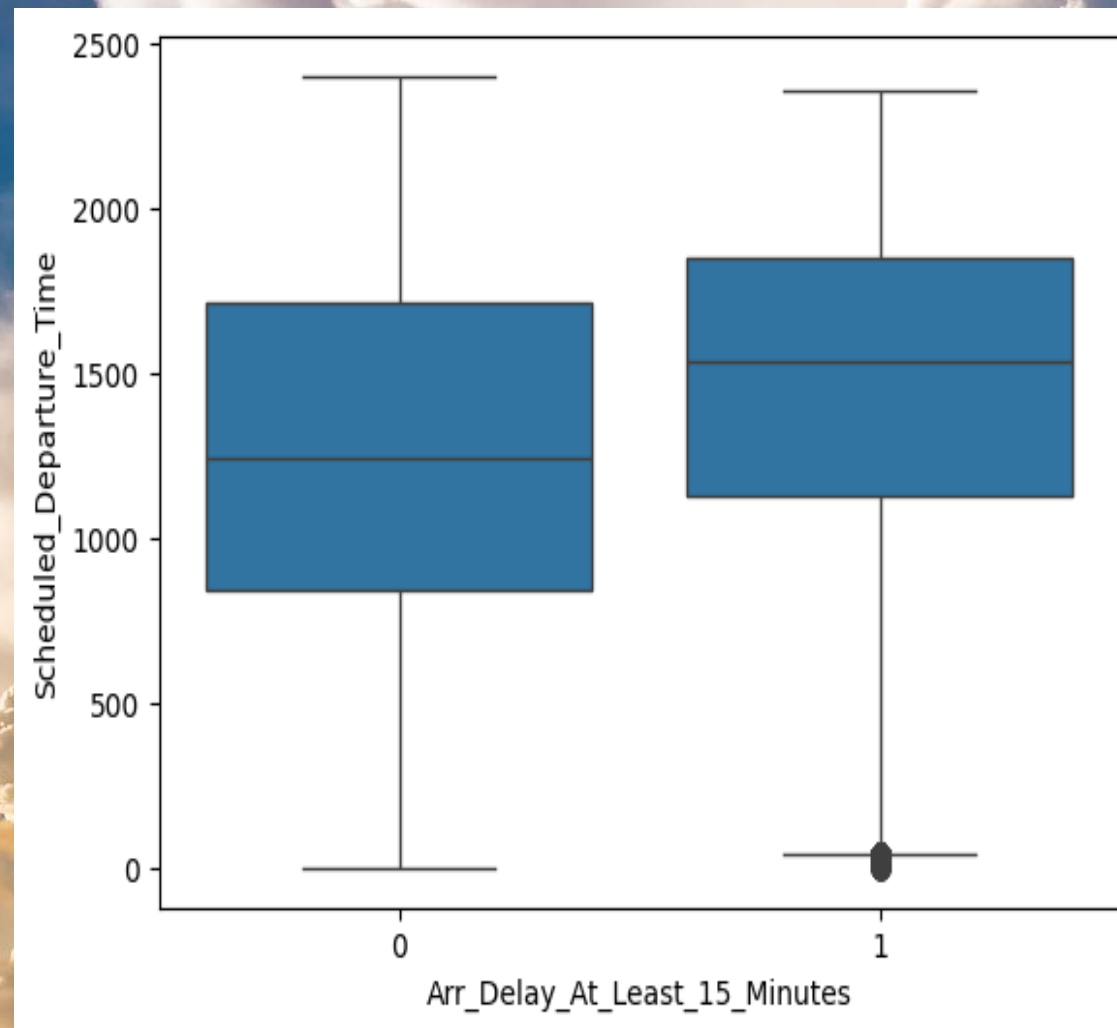
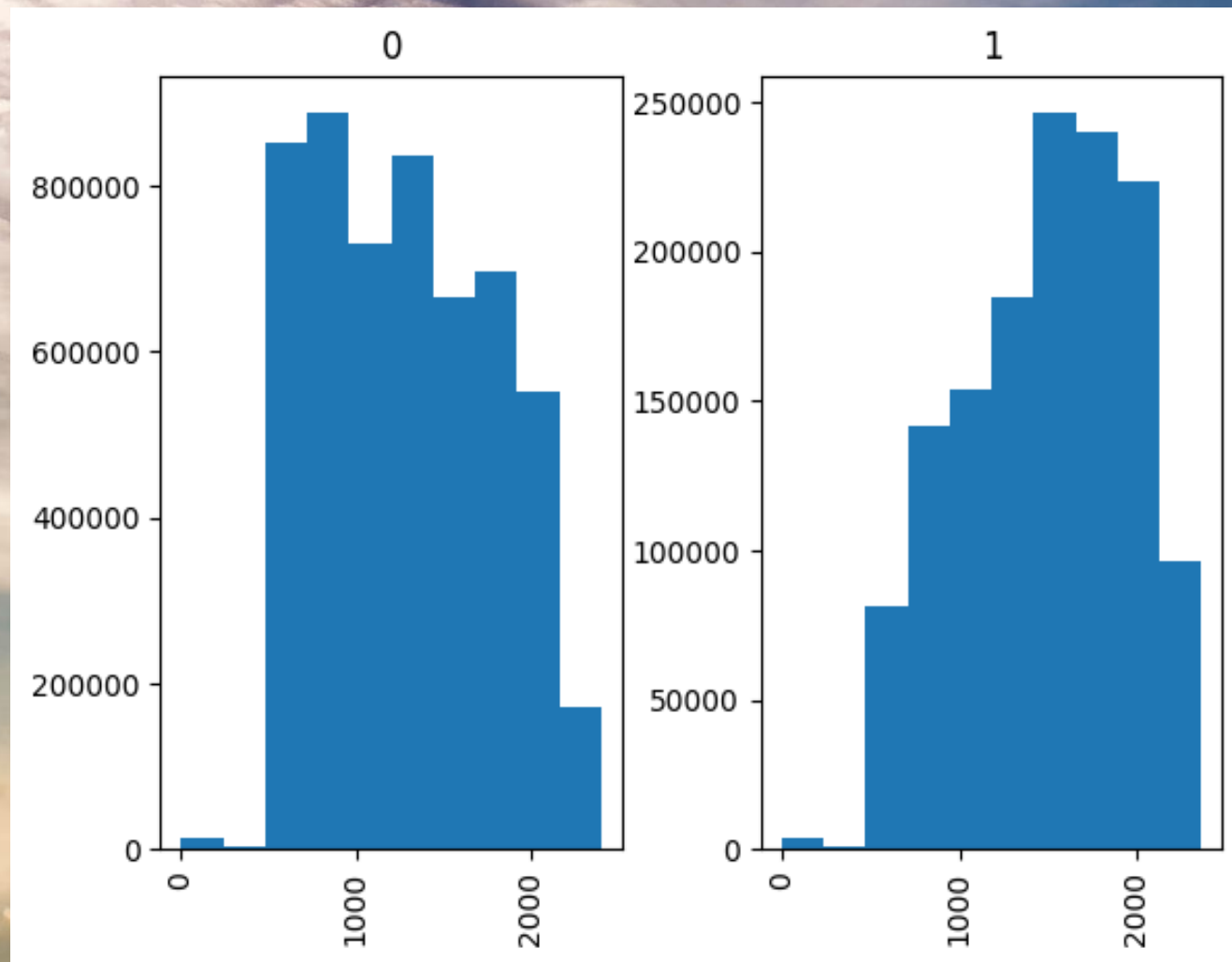


EXPLORATORY DATA ANALYSIS

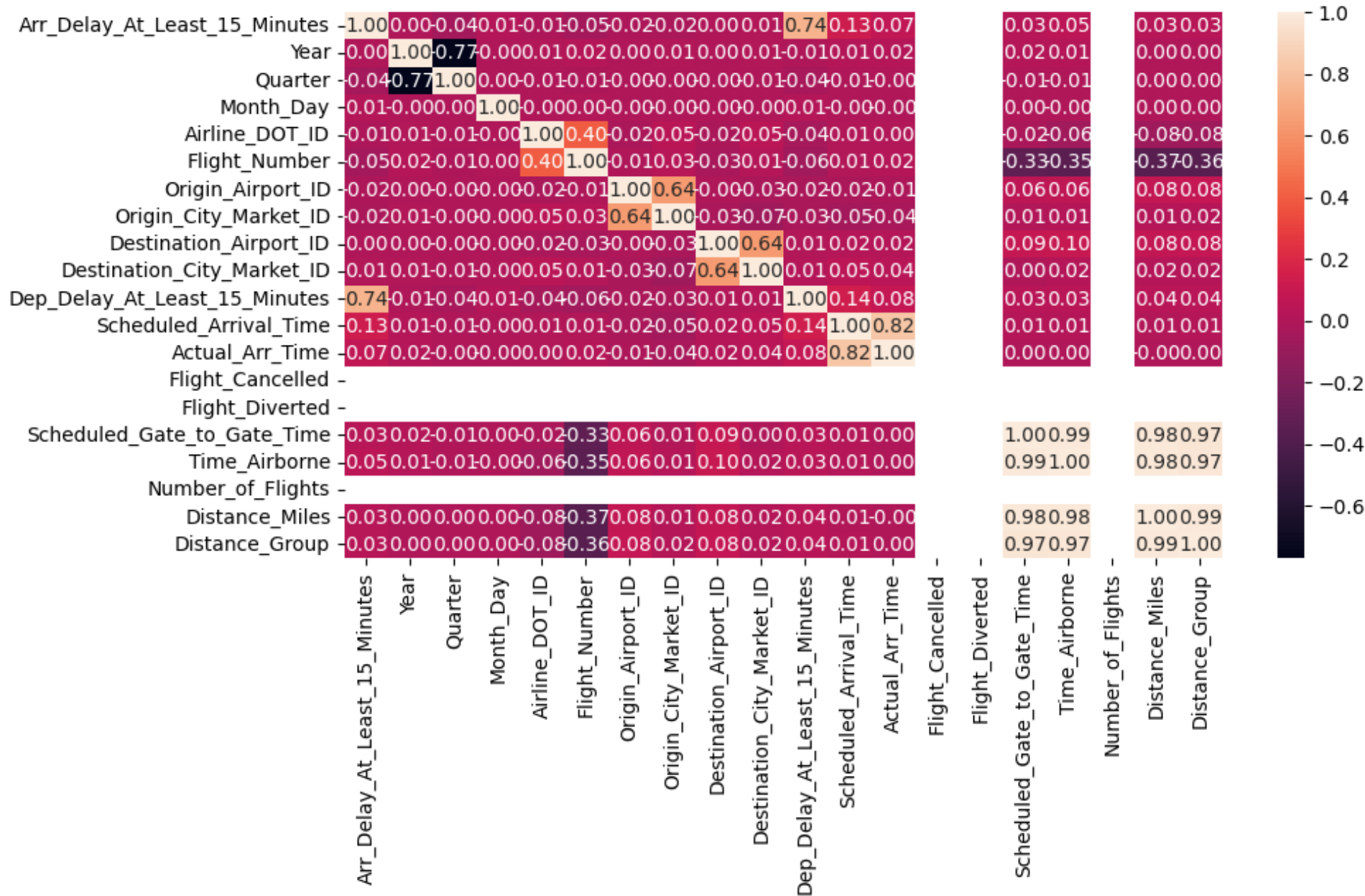
Feature_class	Total_Flights	Delayed_Flights	Percentage_Delayed
5	790,573	185,827	23.51%
7	302,250	67,567	22.35%
4	1,125,895	245,562	21.81%
9	111,202	23,991	21.57%
11	140,112	30,029	21.43%
8	148,064	31,107	21.01%
10	173,193	36,356	20.99%
6	314,425	64,376	20.47%
3	1,357,844	267,365	19.69%
2	1,537,278	281,795	18.33%
1	777,142	137,476	17.69%



EXPLORATORY DATA ANALYSIS



EXPLORATORY DATA ANALYSIS



METHODOLOGY

- Preliminary data analysis
- Explanatory Data Analysis
- Feature engineering
- Fitting model
- Choosing the best model
- Hyper parameters tuning
- Testing Model
- Model Deployment

METHODOLOGY

- Logistic Regression
- Decision Trees
- Random Forest
- Gradient Boosting
- Support Vector Machine
- Naïve Bayes
- K Nearest Neighbor

MODEL COMPARISON

Model	Measure	Downsampled I Dataset
Logistic Regression	Accuracy	60.01%
	Precision	59.34%
	Recall	63.93%
	F1	61.55%
	ROC AUC	66.55%
Decision Tree	Accuracy	59.07%
	Precision	59.96%
	Recall	54.92%
	F1	57.53%
	ROC AUC	59.08%
Random Forest	Accuracy	59.18%
	Precision	59.17%
	Recall	59.57%
	F1	59.57%
	ROC AUC	59.18.00%
Gradient Boosting	Accuracy	61.22%
	Precision	61.18%
	Recall	61.67%
	F1	61.42%
	ROC AUC	61.22%
Naïve Guass	Accuracy	60.20%
	Precision	60.21%
	Recall	60.43%
	F1	60.52%
	ROC AUC	60.20%

MODEL COMPARISON

Model	Measure	Full Dataset	Downsampled l Dataset
Logistic Regression	Accuracy	79.74%	60.01%
	Precision	57.03%	59.34%
	Recall	0.16%	63.93%
	F1	0.33%	61.55%
	ROC AUC	-	66.55%

CONFUSION MATRIX

Original Data Set			
	TRUE		
Predicted	0	1	All
0	2,161,242	702	2,161,944
1	54,316	932	55,248
All	2,215,558	1,634	2,217,192

Downsampled Data Set			
	TRUE		
Predicted	0	1	All
0	164,313	108,667	272,980
1	109,645	165,956	275,601
All	273,958	274,623	548,581

NEXT STEPS

- Feature engineering and hyperparameters tuning in more details
- Creating more meaningful variables
- Exploring different feature combinations and data subset

MODEL DEPLOYMENT

app2 x +

localhost:8504 ☆ | M New Chrome available ⋮

Deploy ⋮

Will your flight be late?

Carrier name

Southwest Airlines Co. ▼

Week day

Monday ▼

Departure Time Block Group

Early Morning ▼

Scheduled Departure Time

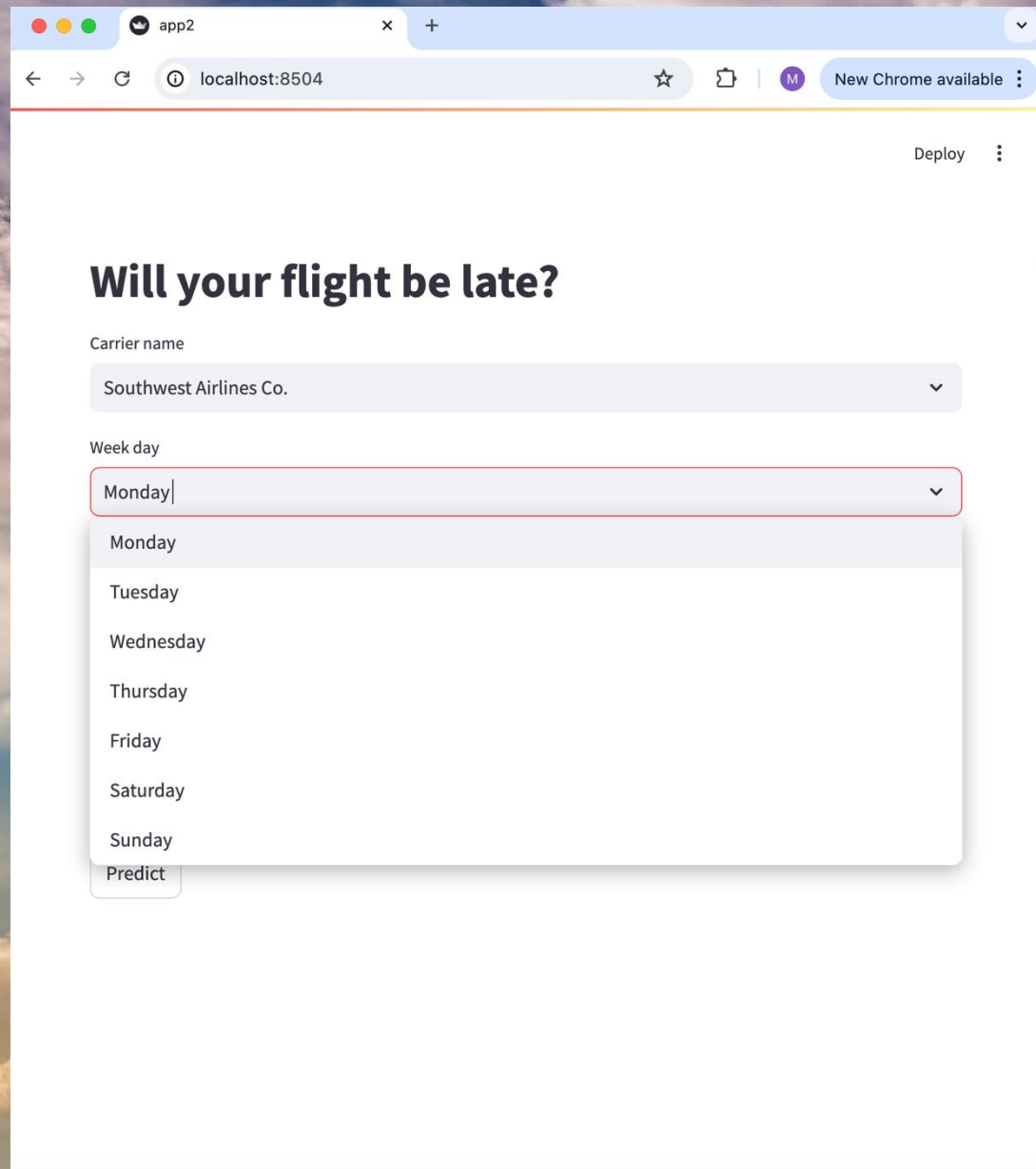
1200 - +

Distance Group

1 - +

Predict

MODEL DEPLOYMENT



app2

localhost:8504

New Chrome available

Deploy

Will your flight be late?

Carrier name

Southwest Airlines Co.

Week day

Monday

Monday

Tuesday

Wednesday

Thursday

Friday

Saturday

Sunday

Predict